Computer Network

Unit 1

- **What is Computer Network**

A computer network is a group of connected devices that communicate with each other to share data and resources. It enables efficient communication and supports services like email, file sharing, and internet access.

Usage of Computer Network

1 **. Resource Sharing**

A computer network allows multiple users to share resources like printers, files, and software.
Example: One printer can be used by many computers.

2**. Communication**

Networks help people communicate easily using email, video calls, and messaging.
Example: Sending emails or chatting online.

3. **Data Sharing**

Data can be transferred from one computer to another through a network.
Example: Sharing documents, images, and videos.

4. Internet Access

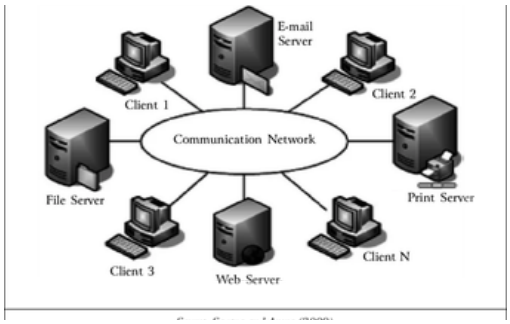
A network provides access to the internet for browsing, learning, and entertainment.
Example: Using Google or watching YouTube.

5. **Online Services**

Networks enable services like online banking, shopping, and ticket booking.
Example: Amazon, online payments.

6. Business Use

Organizations use networks to manage data, communicate, and access databases.
bheading



Advantages of Computer Networks

Here are the advantages uses of a computer network:

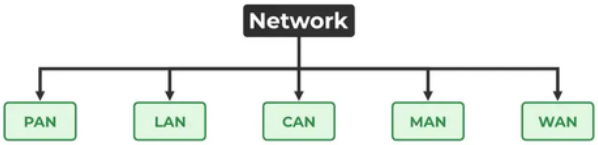
- Improved communication and collaboration.
- Resource sharing may play the role of cost-cutting.
- Data quality management and data security.
- Higher automation benefits and remote access possibilities.
- Enhanced productivity and efficiency

Disadvantages of Computer Networks

Here are the disadvantages uses of a computer network:

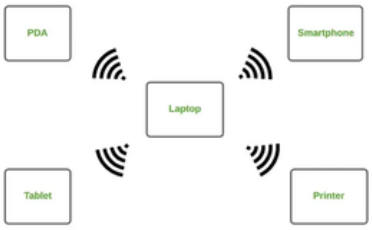
- Data hacking risks and chances of the data leakage.
- The problem of technicalities and operation troubleshooting.
- The likelihood of the exposure of viruses and malware increased.
- There might be a decrease in privacy because of this.

Types of Computer Networks



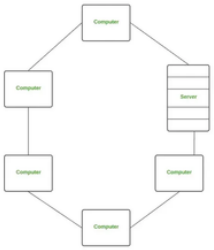
Personal Area Network (PAN)
A Personal Area Network (PAN) connects personal devices within a very short range around a single user.

- **Coverage Range:** 1–10 meters
- **Key Technologies**: Bluetooth, NFC, Infrared
- **Real-World Example:** Smartphone connected to wireless earbuds
- **When It Is Used:** For short-range communication between personal devices

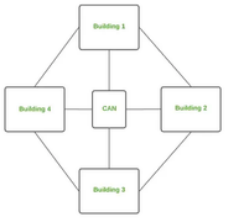


2. Local Area Network (LAN)
A Local Area Network (LAN) connects computers and devices within a limited area such as a home, office, or building.

- **Coverage Range:** One room, building, or small campus
- **Key Technologies:** Ethernet, Wi-Fi
- **Real-World Example:** Home or office Wi-Fi network
- When It Is Used: For fast and secure communication within a local location



3**. Campus Area Network (CAN)**
A Campus Area Network (CAN) connects multiple LANs within a campus or a group of nearby buildings.
Coverage Range: University or corporate campus
Key Technologies: Ethernet, Fiber Optics
Real-World Example: College or corporate campus network
When It Is Used: To interconnect LANs across a campus under one organization



4. Metropolitan Area Network (MAN)
A Metropolitan Area Network (MAN) connects multiple networks within a city
or metropolitan region.

- **Coverage Range:** City or large town
- **Key Technologies:** Fiber Optics, Microwave, Metro Ethernet
- **Real-World Example:** City-wide ISP network
- When It Is Used: To provide high-speed connectivity across a city

5. Wide Area Network (WAN)
A Wide Area Network (WAN) connects networks over large geographical areas such as countries or continents.

- **Coverage Range**: Country, continent, or global
- **Key Technologies:** Leased lines, Satellite links, Internet
- **Real-World Example**: The Internet
- When It Is Used: To enable long-distance and global communication

- **What is Standard Organiztion**

A Standard Organization is an organization that develops rules and standards for data communication in computer networks.

These standards ensure that different devices (computers, mobiles, routers) can communicate properly with each other.

***Standard Organization* হলো এমন সংস্থা *(organization),* যারা *network communication*-এর জন্য *rules (standard)* তৈরি করে।**

এই *rules* মানলে সব *device (computer, mobile, router)* একে অপরের সাথে ঠিকভাবে *communicate* করতে পারে।

1. ISO (International Organization for Standardization)

ISO is an international organization that develops global standards.
In networking, it is famous for the **OSI Model (7 layers).**

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2. IEEE (Institute of Electrical and Electronics Engineers)

IEEE develops standards for electrical, electronics, and networking.
Known for 802 series (LAN & WiFi standards).

Key Work:

- IEEE 802.3 → Ethernet
- IEEE 802.11 → WiFi
- Defines LAN standards

Bengali:

- IEEE একটি সংস্থা, যা electrical, electronics এবং networking-এর standard তৈরি করে
- LAN ও WiFi-এর নিয়ম তৈরি করে

Exam point:
IEEE → LAN & WiFi standard (802 series)

3. IETF (Internet Engineering Task Force)heading

- IETF develops and maintains internet protocols.
- Works through RFC (Request for Comments) documents.

Key Work:

- Develops protocols like TCP, IP, HTTP
- Maintains Internet standards

Bengali:

- IETF হলো এমন একটি সংস্থা, যা Internet-এর protocol তৈরি করে
- TCP/IP, HTTP-এর মতো protocol তৈরি করে

Exam point:
IETF → Internet Protocol develop করে

4. ITU (International Telecommunication Union)

English:

- ITU is a United Nations agency that manages global telecommunication systems.

Key Work:

- Controls radio frequency spectrum
- Satellite communication standards
- Global telecom rules

Bengali:

- ITU হলো United Nations-এর একটি সংস্থা
- এটি global telecommunication system control করে

Exam point:
ITU → Telecommunication & satellite rules

- **What is Protocol?**
  Protocol is a set of rules that governs how data is communicated between devices in a network.
  সহজভাবে বললে (বাংলায়):

Protocol = এমন কিছু নিয়ম, যেগুলো follow করে এক device থেকে অন্য device-এ data পাঠানো হয়।

কেন Protocol দরকার?

যখন দুইটা computer বা mobile একে অপরের সাথে data exchange করে, তখন তাদের ঠিক করতে হয়—

- কিভাবে data পাঠাবে
- কখন পাঠাবে
- কোন format-এ পাঠাবে
- error হলে কী করবে

এই সব কিছুর জন্যই protocol দরকার।

Types of Protocol

1. Application Protocol
   ইউজার সরাসরি ব্যবহার করে
   Example: HTTP, FTP, SMTP
2. Transport Protocol
   ডাটা ঠিকভাবে পাঠানো নিশ্চিত করে
   Example: TCP (reliable), UDP (fast)
3. Network Protocol
   ডাটা কোথায় যাবে ঠিক করে
   Example: IP
4. Security Protocol
   ডাটা নিরাপদ রাখে
   Example: HTTPS, SSL
5. Routing Protocol
   ডাটা কোন পথে যাবে ঠিক করে
   Example: RIP, OSPF

- ## What is Topology

Topology is the arrangement of devices in a computer network and how they are connected to each other**.**

টপোলজি হলো কম্পিউটার নেটওয়ার্কে ডিভাইসগুলো কীভাবে শারীরিক (physical) বা লজিক্যাল (logical)ভাবে সংযুক্ত থাকে তার বিন্যাস।

Types of Topology with Diagram + Advantages & Disadvantages

1 **MESH TOPOLOGY**

Definition

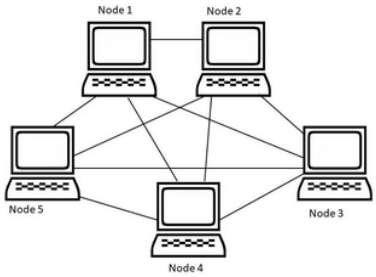
Mesh topology is a network where every device is connected to every other device.

Advantages

- Very reliable
- Fast communication
- High security

Disadvantages

- Very expensive
- Difficult to maintain
- Uses many cables



2. Star Topology

Star topology is a network where all devices are connected to a central device (hub/switch).

Advantages

- Easy to install
- Easy to manage
- One device failure does not affect others

Disadvantages

- Hub failure → whole network stops
- Cost of hub/switch

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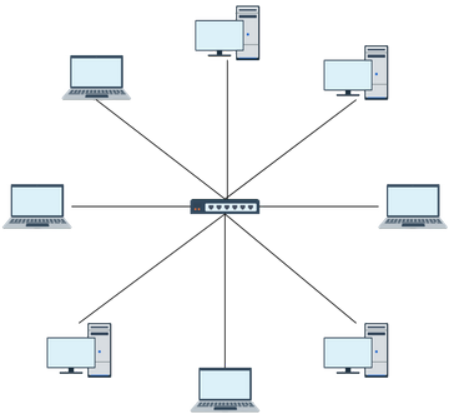


4. Bus Topology

- Bus Topology is a network type in which every computer and network device is connected to a single cable

Advantages of Bus Topology

Low Cost

It is cheap because it uses less cable and no central device like hub or switch.

Easy to Install

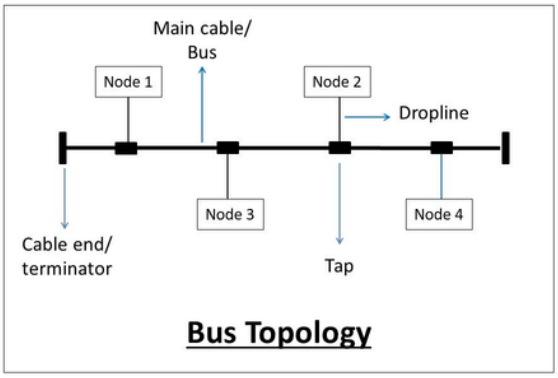
Simple design, so it is easy to set up.

Less Cable Required

Only one main cable is needed.

Good for Small Networks

Works well when there are fewer devices.



- **Disadvantages of Bus Topology**
- **Main Cable Failure**
- If the main cable breaks, the entire network stops.
- **Traffic Problem**
- When many devices send data, network becomes slow.
- **Difficult to Troubleshoot**
- Hard to find where the problem is.
- **Limited Length**
- Cable length is limited, so it cannot cover large areas

- ## Ring Topology

What is Ring Topology?

Ring topology is a network where all devices are connected in a circular loop, and data travels from one device to the next.

Advantages of Ring Topology

No Data Collision

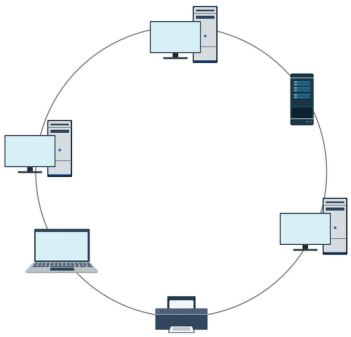
Data moves in one direction (or controlled way), so collisions are avoided.

Equal Access

Every device gets an equal chance to send data.

Better Performance under Load

Works better than bus topology when traffic is high.



Disadvantages of Ring Topology

Single Point Failure

If one device or cable fails, the whole network is affected.

Difficult to Troubleshoot

Hard to find the exact fault location.

Adding/Removing Devices is Difficult

Network must be stopped to make changes.

- ## What is Hybrid Topology?

Hybrid topology is a network that combines two or more different topologies (like star + bus, star + ring).

Advantages of Hybrid Topology

Flexible Design

Can be designed according to needs.

Highly Scalable

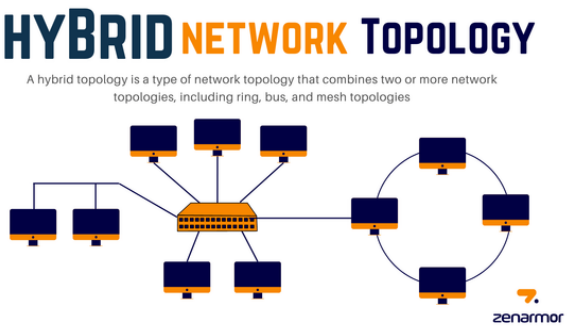
Easy to expand the network.

Reliable

Failure in one part does not affect the whole network.

Better Performance

Combines benefits of different topologies.



Disadvantages of Hybrid Topology

Complex Design
Difficult to design and understand.
High Cost
Requires more devices and cables.
Maintenance is Difficult
Needs skilled management.

- ## Line Configuration –

Line Configuration in Computer Networks

A network is two or more devices connected through a link. A link is a communication pathway that transfers data from one device to another. Devices can be a computer, printer, or any other device that is capable to send and receive data. For visualization purposes, imagine any link as a line drawn between two points
.
For communication to occur, two devices must be connected in some way to the same link at the same time. There are two possible types of connections:

- • **Types of Line Configuration**
- **Point-to-Point Connection**
- **Multipoint Connection**

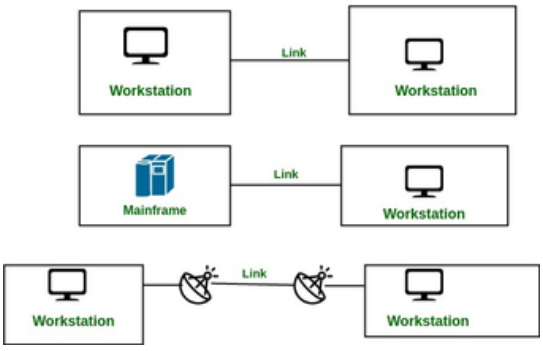
- **Bengali Definition**

লাইন কনফিগারেশন হলো একটি কম্পিউটার নেটওয়ার্কে দুই বা ততোধিক ডিভাইস কীভাবে একটি কমিউনিকেশন লিঙ্ক (যেমন কেবল বা ওয়্যারলেস এর মাধ্যমে সংযুক্ত থাকে তার পদ্ধতি।

- **Point-to-Point Connection:**

1. A point-to-point connection provides a dedicated link between two devices.
2. The entire capacity of the link is reserved for transmission between those two devices.
3. Most point-to-point connections use an actual length of wire or cable to connect the two ends, but other options such as microwave or satellite links are also possible.

4 It is also the simplest to establish and understand.
Example: Point-to-Point connection between the remote control and Television for changing the channels.



- **Advantages of Point-to-Point Connection:**

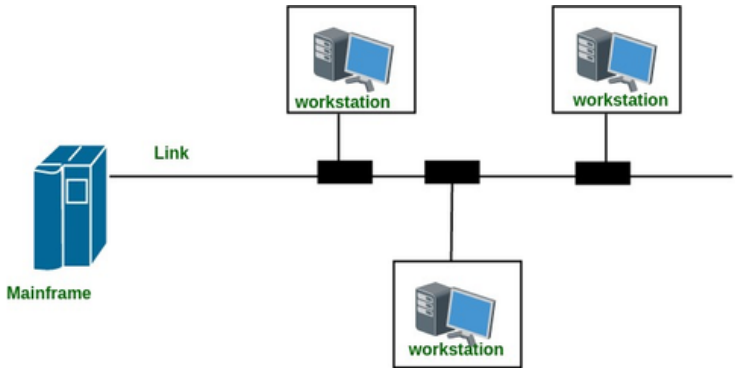
High Bandwidth: A point-to-point connection provides a dedicated link between two devices, which means that the entire capacity of the link is reserved for the two devices. As a result, point-to-point connections usually offer high bandwidth, which makes them suitable for transferring large amounts of data quickly.
Security: Point-to-point connections are more secure than multipoint connections because the link is dedicated to only two devices. There is no risk of other devices eavesdropping on the communication or interfering with it in any way.
R**eliability:** Because a point-to-point connection provides a dedicated link between two devices, it is usually more reliable than a shared link. If there is a problem with the link, it is easier to troubleshoot and fix because there are only two devices involved.
Reduced Latency: Point-to-point connections offer lower latency than multipoint connections because there is no contention for the link. This means that data can be transmitted more quickly, which is especially important for real-time applications such as video conferencing or online gaming.
I**ncreased Contro**l: Point-to-point connections provide greater control over network traffic and bandwidth allocation. This allows for more efficient use of network resources and can help to prevent issues such as network congestion and bottlenecks.

- **Disadvantages of Point-to-Point Connection:**
-
- **Cost:** Setting up a point-to-point connection can be more expensive than setting up a shared link. This is because a dedicated link requires its own infrastructure, such as cables, routers, and switches.
- **Scalability:** Point-to-point connections are not easily scalable, as they require a separate link for each pair of devices. This can be a limitation for large networks that require many devices to be connected.
- **Maintenance:** Because a point-to-point connection provides a dedicated link between two devices, it requires more maintenance than a shared link. If there is a problem with the link, it is the responsibility of the network administrator to troubleshoot and fix it.
- **Limited Flexibility:** Point-to-point connections offer limited flexibility compared to multipoint connections because they only allow communication between two devices. This can be a limitation for applications that require collaboration or coordination between multiple devices.
- **Time-consuming Installation:** Setting up a point-to-point connection can be time-consuming, as it requires a dedicated link to be established between two devices. This can be a disadvantage in situations where time is critical, such as in emergency response situations.

Multipoint Connection :

It is also called Multidrop configuration. In this connection, two or more devices share a single link.
If more than two devices share the link then the channel is considered a 'shared channel'. With shared capacity, there can be two possibilities in a Multipoint Line configuration:

Spatial Sharing: If several devices can share the link simultaneously, it's called Spatially shared line configuration.
Temporal (Time) Sharing: If users must take turns using the link, then it's called Temporally shared or Time Shared Line configuration.



- **Advantages of Multipoint Connection:**

- **Cost**-Effective: Multipoint connections are usually less expensive than point-to-point connections because they allow multiple devices to share the same resources, such as cables, routers, and switches.
- **Scalability:** Multipoint connections are more scalable than point-to-point connections because they allow multiple devices to be connected to the same link. This makes them suitable for large networks that require many devices to be connected.
- **Flexibility:** Multipoint connections are more flexible than point-to-point connections because they allow multiple devices to communicate with each other over the same link. This makes them suitable for applications that require collaboration or coordination between multiple devices.
- I**ncreased Efficiency:** Multipoint connections can improve network efficiency by allowing multiple devices to transmit data simultaneously. This reduces the chances of network congestion and improves overall network performance.
- **Simplified Management:** Multipoint connections simplify network management by reducing the number of physical connections that need to be managed. This reduces the complexity of network administration and makes it easier to troubleshoot problems.

- **Disadvantages of Multipoint Connection:**

- L**imited Bandwidth:** Multipoint connections usually offer limited bandwidth because the link is shared by multiple devices. As a result, the speed of the link may be affected by the number of devices using it.
- **Security:** Multipoint connections are less secure than point-to-point connections because the link is shared by multiple devices. There is a risk of other devices eavesdropping on the communication or interfering with it in some way.
- **Reliability:** Multipoint connections are less reliable than point-to-point connections because the link is shared by multiple devices. If there is a problem with the link, it can affect multiple devices, which can be difficult to troubleshoot and fix.
- L**atency:** Multipoint connections may have higher latency than point-to-point connections due to the increased number of devices accessing the same link. This can lead to delays in data transmission and affect the overall performance of the network.
- **Complexity:** Multipoint connections can be more complex to set up and configure than point-to-point connections. This is because multiple devices need to be connected and configured to work together on the same link, which can be challenging to manage.
- **Network Congestion:** Multipoint connections can be prone to network congestion, especially during peak usage periods. When multiple devices are using the same link simultaneously, it can cause a bottleneck, resulting in slower network speeds and performance issues.

- **<u>Performance of Computer Networkg</u>**

1. **Latency vs Throughput vs Bandwidth**

- • **Bandwidth:** This is about the volume of data that can be transferred over a network. The standard measurement for data transfer speed is megabits per second (Mbps).

• **Throughput**: While bandwidth tells you how much data could theoretically be transferred across the network, throughput indicates how much data was actually transferred from a source at any given time.

- • **Latency:** This measures the time it takes for data to get to its destination across the network. It is commonly measured as a round trip delay. Latency is usually measured in milliseconds (ms).

Latency vs Throughput vs Bandwidth
• Bandwidth: এটি বোঝায় একটি নেটওয়ার্কের মাধ্যমে কত পরিমাণ ডেটা পাঠানো যায়। ডেটা ট্রান্সফারের গতি সাধারণত megabits per second (Mbps) এ মাপা হয়।
• Throughput: Bandwidth যেখানে বলে কত ডেটা তাত্ত্বিকভাবে পাঠানো সম্ভব, সেখানে Throughput দেখায় আসলে নির্দিষ্ট সময়ে কত ডেটা সত্যিই পাঠানো হয়েছে।
• Latency: এটি মাপে একটি ডেটা তার গন্তব্যে পৌঁছাতে কত সময় লাগে। এটি সাধারণত round trip delay হিসেবে মাপা হয়। Latency সাধারণত milliseconds (ms) এ মাপা হয়।

- **Bandwidth & Throughput Unit (Table)**
-
- **Bandwidth bps - bits per second Base unit**
- **Kbps - Kilobits per second 1 Kbps = 1,000 bps**
- **Mbps - Megabits per second 1 Mbps = 1,000,000 bps**
- **Gbps - Gigabits per second 1 Gbps = 1,000,000,000 bps**

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- **Throughput bps bits per second Base unit**
- **Kbps - Kilobits per second 1 Kbps = 1,000 bps**
- **Mbps - Megabits per second 1 Mbps = 1,000,000 bps**
- **Gbps - Gigabits per second 1 Gbps = 1,000,000,000 bps**

3. Bit vs Byte
➤ Relation:
1 Byte (B) = 8 bits (b)

Advantages of Bandwidth

- Capacity Planning: Helps in designing and planning network infrastructure by defining the maximum potential capacity.
- Performance Benchmarking: Provides a standard measure to compare different networks and channels.

Disadvantages of Bandwidth

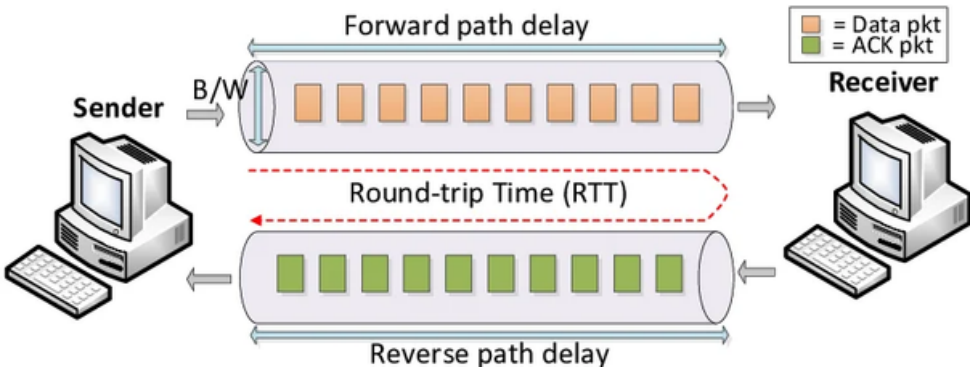
- Theoretical Limit: It is a theoretical measure and does not account for real-world factors like network congestion.
- Doesn't Reflect Real Performance: High bandwidth doesn’t always translate to high actual data transfer rates due to various inefficiencies.

Advantages of Throughput

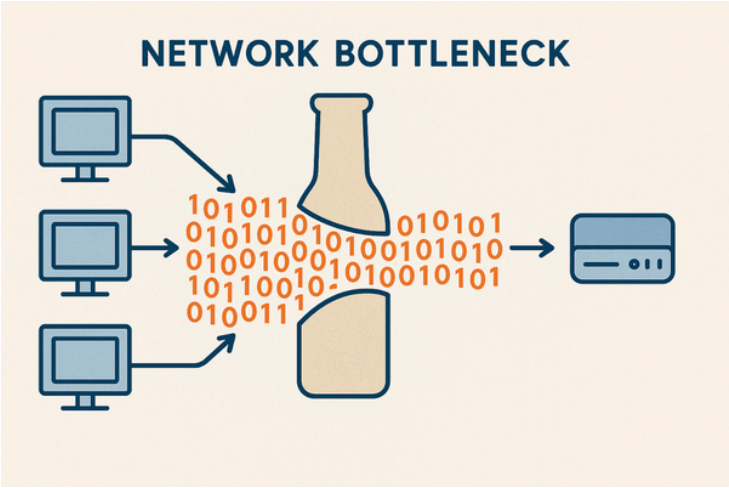
- Realistic Measurement: Provides a practical measure of how much data is being transmitted, taking into account network conditions and overheads.
- Performance Evaluation: Useful for assessing the actual performance and efficiency of the network.

Disadvantages of Throughput

- Variable Rates: Can vary due to network congestion, errors, and other factors, making it less consistent.
- Affected by Latency: Throughput is influenced by latency, which can impact the rate of data transfer.



BANDWIDHT DIAGRAM

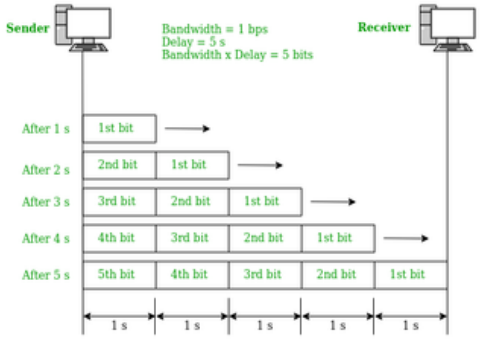


THROUGHPUT DIAGRAM

- **BANDWIDTH - DELAY PRODUCT**

Bandwidth and Delay are two performance measurements of a link. However, what is significant in data communications is the product of the two, the bandwidth-delay product. Let us take two hypothetical cases as examples.

Case 1: Assume a link is of bandwidth 1bps and the delay of the link is 5s. Let us find the bandwidth-delay product in this case. From the image, we can say that this product 1 x 5 is the maximum number of bits that can fill the link. There can be close to 5 bits at any time on the link.

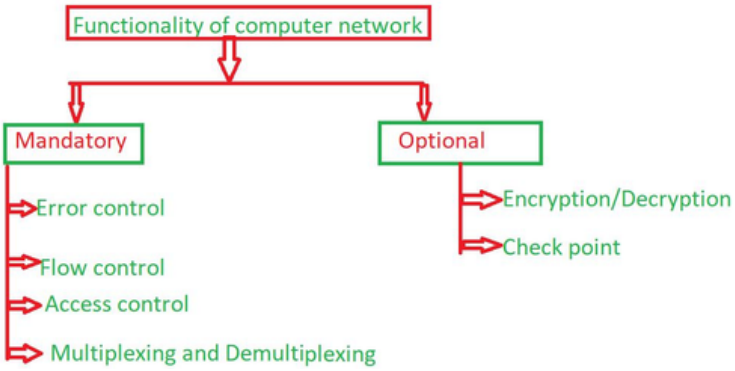


JITTER

Jitter is another performance issue related to the delay. In technical terms, jitter is a "packet delay variance". It can simply mean that jitter is considered a problem when different packets of data face different delays in a network and the data at the receiver application is time-sensitive, i.e. audio or video data. Jitter is measured in milliseconds(ms). It is defined as an interference in the normal order of sending data packets. For example: if the delay for the first packet is 10 ms, for the second is 35 ms, and for the third is 50 ms, then the real-time destination application that uses the packets experiences jitter.

- **Functionality of Computer Network:**
- **Computer network has two type of functionality:**

1. **Mandatory function**
2. **Optional function**



1. **Mandatory function**: Some function is mandatory in computer network and when mandatory function include in computer network, then perform operation in computer network.Some mandatory functions are given below:

- **Error control:** The computer network has some responsibility like transmission of data from one device to another device and end to end transfer of data from a transmitting application to a receiving application involves many steps, each subject to error. By using the error control process, we can be confident that the transmitted and received data are identical. Data can be corrupted during transmission. The error must be detected and corrected for reliable communication. Types of error-
- (**a). Single bit error:** The terms single bit error means that only one bit of the data unit was changed from 1 to 0 and 0 to 1.
- **(b). Burst Error**: The term burst error means that two or more bits in the data unit were changed. A burst error is also called packet-level error, where errors like packet loss, duplication, reordering.
- **Flow control:** When a packet (Layer-2 data) is sent from one host to another over a single medium, it is required that the sender and receiver should work at the same speed. That is, the sender sends at a speed on which the receiver can process and accept the data. If the sender is sending too fast the receiver may be overloaded, (swamped) and data may be lost.
- **Access control:** Network access control is a method of enhancing the security of a private organizational network by restricting the availability of network resources to endpoint devices that comply with the organization's security policy. The network access control scheme comprises of two major components such as Restricted Access and Network Boundary Protection.
- **Multiplexing and Demultiplexing:** A multiplexing is a technique by which different analog and digital streams of transmission can be simultaneously processed over a shared link. Multiplexing divides the high capacity medium into low capacity logical medium which is then shared by different streams.

2. Optional function: Some function is optional in computer network and optional function do not need always in computer network.Some optional functions are given below:

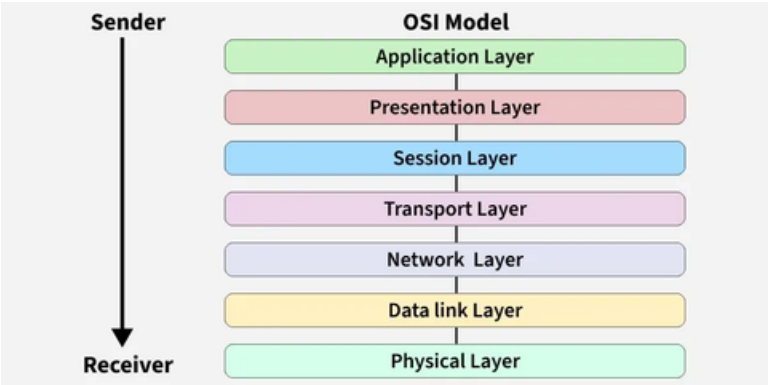
- **Encryption and decryption:** Decryption and Encryption is a security method in which information is encoded in such a way that only authorized user can read it. Some network uses encryption algorithm to generate ciphertext that can only be read if decrypted. Types of encryption-

(**a). Symmetric Key encryption**
(**b). Public Key encryption**

- **Check Points:** Let us say you are downloading a file of 500mb and suppose a failure occur and after some time if you re-start that download then it should resume from where failure happend. This is known as check-point creation. You can create multiple check points, after failure it will start from its predecessor check-point. Check-points are optional functionality of computer network. It is not mandatory to follow by all networks. It is usefull in downloading and uploading large files

What is OSI Model? - Layers of OSI Model

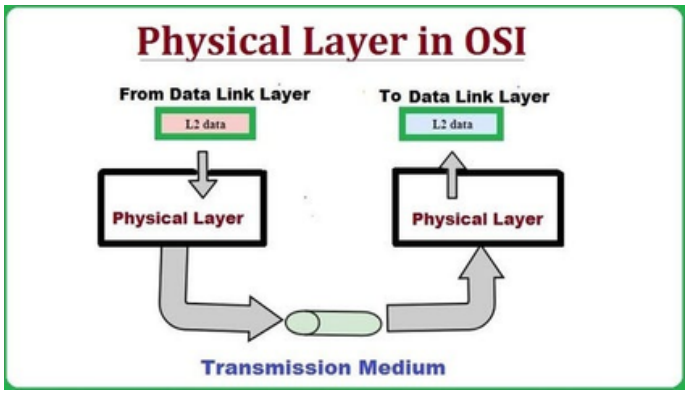
The OSI (Open Systems Interconnection) Model is a set of rules that explains how different computer systems communicate over a network. OSI Model was developed by the International Organization for Standardization (ISO). The OSI Model consists of 7 layers and each layer has specific functions and responsibilities. This layered approach makes it easier for different devices and technologies to work together.



- **Layer 1: Physical Layer**

The lowest layer of the OSI reference model is the Physical Layer. It is responsible for the actual physical connection between the devices. The physical layer contains information in the form of bits.

- **Physical Layer** is responsible for transmitting individual bits from one node to the next.
- When receiving data, this layer will get the signal received and convert it into 0s and 1s and send them to the Data Link layer, which will put the frame back together.
- Common physical layer devices are **Hub**, **Repeater**, **Modem,** and **Cables**.



- **Cables and Connectors:**

The physical layer defines the cables (like twisted pair, coaxial, and fiber optics) and connectors (such as RJ45) used to link devices. These ensure signals can pass from one device to another reliably and efficiently.

- **Physical Topology:**

The physical layer determines how devices are physically arranged—whether in a bus, star, ring, or mesh topology. It defines how data paths are laid out physically, impacting how devices communicate.

- **Repeaters and Hubs (Hardware):**

A repeater is a device that regenerates and amplifies weak signals to extend the distance a signal can travel without degradation. In contrast, a hub is a multi-port device that connects multiple devices in a network. It broadcasts any incoming signal to all ports, meaning every device connected to the hub receives the same data simultaneously. Neither repeaters nor hubs perform any data filtering or intelligent routing.

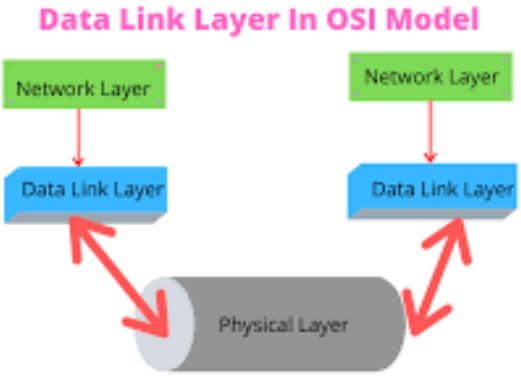
- **Transmission (Simplex, Half-Duplex, Full-Duplex):**
- In simplex transmission, data flows in one direction only, like a TV broadcast.
- Half-duplex allows data to flow in both directions, but only one direction at a time (like a walkie-talkie).
- Full-duplex allows simultaneous two-way communication—both sides can send and receive data at the same time, like a phone call.

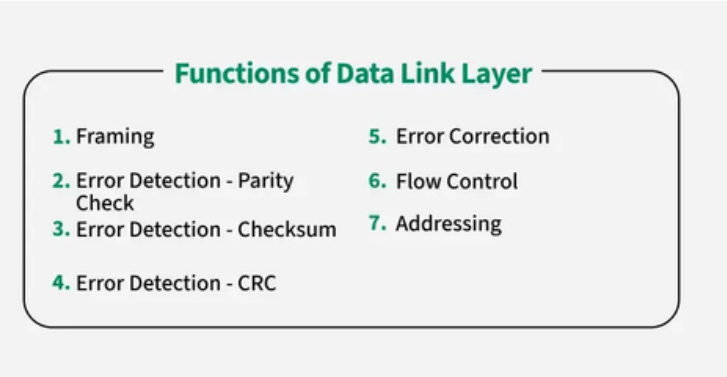
- **Encoding:**

Encoding is the process of converting digital data into signals suitable for transmission. For example, in Non-Return-to-Zero (NRZ) encoding, bits are represented by different voltage levels, while in Manchester encoding, each bit has a transition to ensure synchronization. Encoding ensures reliable interpretation of signals.

- ## Data Link Layer Services

Data Link Layer is the second layer of seven-layer Open System Interconnection (OSI) reference model of computer networking and lies just above Physical Layer. It is responsible for receiving and getting data bits usually from Physical Layer and then converting these bits into groups, known as data link frames so that it can be transmitted further. It is also responsible to handle errors that might arise due to transmission of bits.





1. Hop hop to Delivery

The data link layer moves data from one directly connected node to the next. Its job is to take packets from the network layer and break them into frames for transmission over the physical link. It also handles error detection and correction on that link, ensuring each hop's data is reliable before passing it onward.

2. Error Detection

To detect errors, some common methods include:

1. Simple Parity Check :

- It involves adding an extra bit (called the parity bit) to a string of data bits.
- This bit is set to make the total number of 1's either even (even parity) or odd (odd parity).
- When the data is received or read, the parity bit is checked to see if it matches the expected value (even or odd).
- If it doesn't, it indicates that an error may have occurred in the data transmission.

- **. Checksum:**
- The sum of data bits is calculated and a small value (the checksum) is appended to the data.
- The receiver performs the same calculation and compares the result to the transmitted checksum.
- If they match, the data is considered valid and if not, an error is detected.
-

- **CRC (Cyclic Redundancy Check):**
- It treats data as a large polynomial and divides it by a predefined polynomial divisor.
- The remainder of this division is appended to the data.
- The receiver performs the same division and checks if the remainder matches the transmitted CRC.
- If there's a mismatch, an error is detected.

3. Flow Control

It is a method to manage the rate of data transmission between sender and receiver in order to prevent buffer overflow or congestion. Flow control methods include:

- I**n Stop-and-Wait,** the sender transmits one frame at a time and waits for an acknowledgment (ACK) from the receiver before sending the next frame.
- I**n Go-Back-N ARQ**, the sender can send multiple frames, but if one is lost or has error, all subsequent frames must be retransmitted.
- **In Selective Repeat ARQ,** only the frames with errors or lost frames are retransmitted, improving efficiency.

4. **Physical Addressing**

Addressing at the Data Link Layer is used for identifying devices within a local network or link. It uses physical or MAC (Media Access Control) addresses to ensure that data is delivered to the correct device.

MAC Address is a hardware address associated to network interface card (NIC) or network device. The MAC address is used to route frames between devices within the same local network or between connected networks in a point-to-point link.

Logical addressing can also be used with MAC addressing to provide higher-level services such as multiplexing or identifying different protocols (Ethernet, Wi-Fi, etc.).

5. **Frame (Data Link Layer)**

- A frame is the unit of data at the data link layer.
- It is created by adding a header (MAC address) and a trailer (error checking like CRC) to a packet.
- Frames are used for node-to-node (hop-to-hop) delivery within a local network.
- It ensures the data is correctly delivered between directly connected devices.

6. **Packet (Network Layer)**

- A packet is the unit of data at the network layer.
- It contains IP addresses and carries data from the source to the destination across multiple networks.
- Packets are responsible for end-to-end delivery over the internet.
- They are later converted into frames for transmission on each link.

Types of Ethernet Cable

An ethernet cable allows the user to connect their devices such as computers, mobile phones, routers, etc, to a Local Area Network (LAN) that will allow a user to have internet access, and able to communicate with each other through a wired connection. It also carries broadband signals between devices connected through it. In this article, we are going to discuss different types of ethernet cable used in local area networks for reliable internet connection.

- **Types of Ethernet Cables**

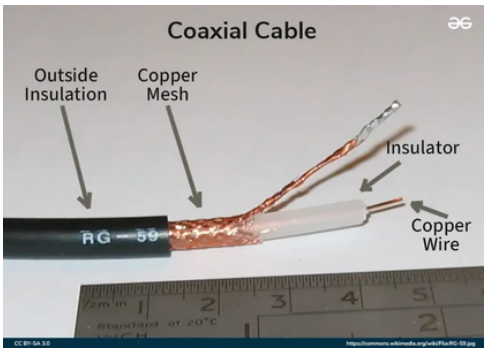
Mainly there are three types of ethernet cables used in LANs

- Coaxial Cables
- Twisted Pair Cables
- Fiber optic Cables

1. Coaxial Cables

A coaxial cable is used to carry high-frequency electrical signals with low losses. It uses 10Base2 and 10Base5 Ethernet variants. It has a copper conductor in the middle that is surrounded by a dielectric insulator generally made of PVC or Teflon. The dielectric insulator is surrounded by a plaited conducting metallic shield which reduces Electromagnetic Interference of the metal and outside interference and finally, the metallic shield is covered by a plastic covering called a sheath usually made of PVC or some other fire-resistant plastic material. Its maximum transmission speed is 10 Mbps. It is usually used in telephone systems, cable TV, etc.

- **Design:** Coaxial cable design choices affect physical size, frequency performance, attenuation, power handling capabilities, flexibility, strength, and cost. It consists of an inner conductor which might be solid or stranded surrounded by an insulator and, to provide flexibility, it is further surrounded by a copper mesh which is further surrounded by a plastic or insulating jacket.



Types of Coaxial Cables

- **Hardline Coaxial Cable:** It is used in applications where high signal strength is required, this type is most commonly used. They are used in internet lines and telephone lines.
- **RG-6 Coaxial Cable:** It is used where better signal quality is required; it has a thicker dielectric insulator, they are used in broadband internet, cable TV, etc.
- **Tri-Axial Cable:** They offer more bandwidth and interference rejection; they use an additional copper braid shield. Commonly used in connecting cameras and cable TVs. Etc.

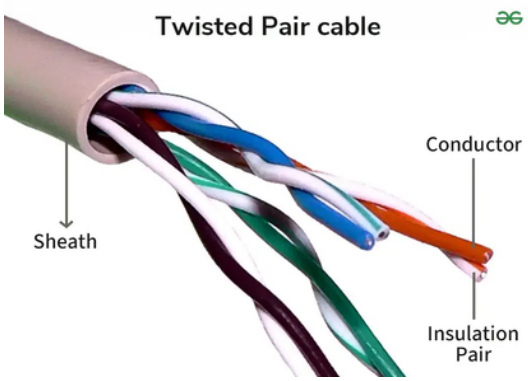
Types of Connectors Used in Coaxial Cable

- BNC (Bayonet Neil Concelman)
- N series Connectors
- F Type Connectors
- SMA or Subminiature Connector
- TNC (Threaded Neil Concelman) etc.

- **Twisted Pair Cable**

A twisted pair is a copper wire cable in which two insulated copper wires are twisted around each other to reduce interference or crosstalk. It uses 10BASE-T, 100BASE-T, and some other newer ethernet variants. It uses RJ-45 connectors.

Design: A twisted pair cable usually contains two or more conducting wires either shielded by an insulator or not and, further these twisted pairs of wires are coated for protection from any damage.



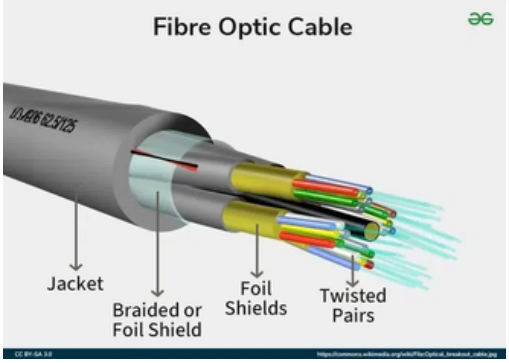
Types of Twisted Pair Cable

- **Shielded Twisted Pair (STP) Cable: In** STP the wires are covered by a copper braid covering or a foil shield, this foil shield adds a layer that protects it against interference leaking into and out of the cable. Hence, they are used for longer distances and higher transmission rates.
- **Unshielded Twisted Pair (UTP) Cable:** Unshielded twisted pair cable is one of the most commonly used cables in computer networks at present time. UTP consists of two insulated copper wires twisted around one another, the twisting of wires helps in controlling interference.

- **Fiber Optic Cable**

Fiber optic cables use optical fibers which are made of glass cores surrounded by several layers of covering material generally made of PVC or Teflon. It transmits data in the form of light signals due to which there are no interference issues in fiber optics. Fiber optics can transmit signals over a very long distance as compared to twisted pairs or coaxial cables. It uses 10BaseF, 100BaseFX, 100BaseBX, 100BaseSX, 1000BaseFx, 1000BaseSX, and 1000BaseBx ethernet variants. Hence, it is capable of carrying information at a great speed.

- **Design:** An optical fiber consists of a core and a cladding chosen for their total internal reflection due to the difference in refractive index between the two. In real optical fibers, the cladding is usually covered with a layer of acrylate or polyimide polymer. The coating protects the fiber from damage and several layers of protective sheathing, depending on the application are added to form the cable.



Types of Connectors Used in Fiber Optics Cable

- ST (Straight-Tip) Connector
- FC (Fiber Channel) Connector
- SC (Subscriber) Connector
- LC (Lucent) Connector